

DISCELL

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Abstract

[TODO: abstract]

1 Introduction

Imaging-based spatial transcriptomics measures thousands of genes in every cell of a tissue section, in place (Janesick et al., 2023), so a cell’s expression can be read together with the cells around it. The counts assigned to a cell, however, come from three sources. The first is the cell’s *intrinsic state*: its type, its cell-cycle phase and whatever else it would carry anywhere in the tissue. The second is its *response* to its niche: programmes that change with who its neighbours are and what tissue surrounds it. The third is *spill-over*: transcripts of neighbouring cells assigned to it, because segmentation of densely packed tissue is imperfect. On this platform a typical cell carries about a tenth of its counts from neighbours (Yang et al., 2025).

The response and spill-over are confounded at first order. A cell that receives its neighbours’ transcripts acquires their markers; a cell that responds to its neighbours shifts its programmes in a way that their types predict. Both grow with the share of a neighbouring type (Bilous et al., 2026), and spill-over alone produces apparent cell-cell interaction signals (Mitchel et al., 2026). Where the responding genes are also markers of the neighbours, as for many ligands and receptors, the two leave the same signature. They differ in form (one adds a neighbour’s profile, the other shifts the cell’s own programmes), but from the counts alone the split between them is set by the modelling assumptions, and a flexible decoder can mimic either. Every claim that a spatial effect on expression is real therefore rests, usually silently, on how much of the neighbour-predictable signal was attributed to spill-over.

Related work. Classical models split a gene’s variance into spatial, intrinsic and environmental parts (Svensson et al., 2018; Arnol et al., 2019) or regress expression on niche covariates (Cable et al., 2022; Tanevski et al., 2022), and treat contamination separately, before analysis (Young and Behjati, 2020; Fleming et al., 2023; Petukhov et al., 2022; Mitchel et al., 2026). Among generative models of the cell, each models only one side of the confound (Table 1). On the spill-over side, resolVI (Ergen and Yosef, 2026) writes a cell’s counts as a mixture of its own decoded expression, its neighbours’ expression decoded by the same network, and a background, with the mixing weights estimated per cell. It has no niche term in the cell’s own expression, and it separates the components by expecting true expression to be low-dimensional. A niche response is low-dimensional too, and the part of it that moves a cell toward its neigh-

Table 1: The closest generative models, on the properties this paper turns on. *Spill-over*: misassigned transcripts are part of the generative model. *Invariance*: the regulariser that keeps the intrinsic latent free of the niche. *Intervention*: expression generated under an explicit change of the neighbourhood. ✓ yes, × no. ^aSIMVI instead estimates a spatial treatment effect.

Model	Niche	Spill-over	Invariance	Intervention
resolVI	×	estimated per cell	×	×
SIMVI	✓	×	independence, marginal adversarial, given type	× ^a
MintFlow	✓	×	adversarial, marginal adversarial, given type	delete, replace cells
Cellina	✓	×	adversarial, marginal adversarial, given type	swap neighbours
DISCELL	✓	one fraction, swept	adversarial, given type	relocate a type

bours’ profile acts like spill-over, so that expectation alone does not separate the two. Our spill-over term is resolVI’s neighbour mixture with the weights fixed. On the niche side, NCEM (Fischer et al., 2023) regresses a cell’s expression on its neighbourhood given cell-type labels, including a conditional-VAE variant with a per-cell latent. SIMVI (Dong et al., 2025) splits a cell into an intrinsic latent and a spatially induced one, whose prior is a regression on the neighbours’ intrinsic latents, the closest analogue of our response prior, with an independence penalty applied to the intrinsic latent. MintFlow (Akbarnejad et al., 2025) conditions the intrinsic latent on cell type and keeps neighbour-type composition out of it with type-specific adversarial critics, the type-conditional invariance we also use. Cellina (Moeed et al., 2026) encodes an extrinsic latent from the neighbours’ expression, removes spatial-domain labels from the intrinsic latent adversarially, and predicts expression under a swapped neighbourhood. Celcomen (Megas et al., 2026) and NicheCompass (Birk et al., 2025) model inter-cellular gene programmes without a per-cell split. None of these models spill-over, and MintFlow and Cellina name segmentation errors as a limitation. SIMVI and MintFlow state identifiability results for their latents, but no method states what the counts cannot identify when a response and spill-over compete for them.

Our approach. DISCELL models both sides. A cell’s expected composition is a mixture of its own expression, built from an intrinsic state $\mathbf{z}_i$ and a response $\mathbf{w}_i$ whose prior depends on a descriptor of

the niche, and a fraction κ of its neighbours’ expression (Section 2.1). The counts bound κ from above but never from below (Proposition 1), so we do not estimate it. We refit the model over a grid of κ and report, for each finding, its *breakdown point* κ^* : the smallest spill-over fraction that explains the finding away (Definition 1). This is a sensitivity analysis in the tradition of tipping points and partial identification (Rosenbaum, 2002; Manski, 2003), with κ as the sensitivity parameter. Neither ingredient is new alone: the spill-over term is resolVI’s mixture with fixed weights, and adversarial invariance given type is MintFlow’s. What is new is putting a niche response and spill-over into one model, which creates the confound, and the sweep, which handles it.

Contributions.

- **A generative model in which a niche response and spill-over compete for the same counts** (Section 2.1).
- **Conditional invariance with a held-out probe, as an evaluation protocol.** A type-conditional adversary keeps neighbour composition and an image-derived niche descriptor out of $\mathbf{z}$; a probe trained after the fit measures the residual niche signal against a within-type permutation floor, for every method on the same cells (Section 2.3).
- **A spill-over sweep with breakdown points**, transferable to any model with a spill-over term, and with teeth: **[PENDING RUNS: breakdown-gaps queue: it breaks three findings, while the others hold to $\kappa = 0.4$, four times the typical misassignment]** (Section 2.4).
- **An evaluation on four sections** of ovarian cancer, lung cancer and paediatric lung, plus a serial section held out whole, against resolVI, SIMVI, MintFlow and Cellina, with recovery checks on simulated sections (Section 3). Code: **[URL to be added]**.

2 Model

DISCELL describes the counts of each segmented cell as the sum of three parts: what the cell is (its *intrinsic state*), how it shifts with its surroundings (its *response*), and transcripts that belong to its neighbours (*spill-over*). The first two are latent variables of the cell; the third is a fixed mixture with the neighbours’ expression whose size, the spill-over fraction κ , cannot be learned from the counts and is therefore swept (Figure 1).

2.1 Generative model

For each of N segmented cells i on one section we observe a count vector $\mathbf{x}_i$ over G genes with total ℓ_i , a lineage label $t_i \in \{1, \dots, K\}$, its position and the morphology image around it. Two cells are neighbours if their Voronoi cells share a face (Delaunay adjacency), with edges longer than $40 \mu\text{m}$ pruned; $\mathcal{N}_i$ is the neighbour set of i . The model is

$$\mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (1)$$

$$\mathbf{w}_i \mid \mathbf{c}_i, t_i \sim \mathcal{N}(m_\psi(\mathbf{c}_i, t_i), \sigma_w^2 \mathbf{I}), \quad (2)$$

$$\boldsymbol{\rho}_i = \text{softmax} \left(\underbrace{a(\mathbf{z}_i)}_{\text{intrinsic}} + \underbrace{\mathbf{B}\mathbf{w}_i}_{\text{response}} \right), \quad (3)$$

$$\bar{\boldsymbol{\rho}}_i = \sum_{j \in \mathcal{N}_i} \beta_{ij} \boldsymbol{\rho}_j, \quad (4)$$

$$\mathbf{p}_i = (1 - \kappa) \boldsymbol{\rho}_i + \underbrace{\kappa \bar{\boldsymbol{\rho}}_i}_{\text{spill-over}}, \quad (5)$$

$$\mathbf{x}_i \mid \ell_i \sim \text{Multinomial}(\ell_i, \mathbf{p}_i). \quad (6)$$

Here $\boldsymbol{\rho}_i$ is the cell’s own, *clean* composition over genes. Its intrinsic part $a(\mathbf{z}_i)$ is a decoder of the intrinsic state $\mathbf{z}_i \in \mathbb{R}^{d_z}$; its response part is linear in the response $\mathbf{w}_i \in \mathbb{R}^{d_w}$, so that column k of the loadings $\mathbf{B}$ is a gene programme and w_{ik} is how far the cell has moved along it. The decoder has no type input: the only route from identity to expression runs through $\mathbf{z}_i$, which therefore carries the type together with what the label discretises away, such as cell-cycle phase. The prior mean $m_\psi(\mathbf{c}_i, t_i)$ is the response expected of a cell of type t_i in niche $\mathbf{c}_i$; the prior scale is fixed at $\sigma_w = 1$, which pins the scale of $\mathbf{w}$ against that of $\mathbf{B}$.

The niche descriptor. $\mathbf{c}_i$ is a deterministic function of the data, shared by co-located cells and containing no latent:

$$\mathbf{c}_i = \text{GATv2}(\text{embed}(t_i); \{t_j\}_{j \in \mathcal{N}_i}) \oplus \Phi_i \oplus \mathbf{1}[\mathcal{N}_i = \emptyset]. \quad (7)$$

The first part is one graph-attention layer (Brody et al., 2022) over the neighbours’ types, queried with the cell’s own type, which amounts to a learned, type-specific reweighting of the neighbour-type composition $\mathbf{y}_i$. The second, Φ_i , is the embedding by a frozen pretrained image model (Shaban et al., 2025) of a $128 \mu\text{m}$ morphology crop around the cell, in which a $25 \mu\text{m}$ disc over the cell is set to zero, so that it describes the surrounding tissue and not the cell. The query is the type rather than $\mathbf{z}_i$, and no distances enter, for two reasons: a $\mathbf{z}$ -query would let the prior see the cell itself, and distances would let the attention rediscover the spill-over kernel and merge the two mechanisms the model separates (Supplement S2.2).

Figure 1: DISCELL in four panels. **a**, Spill-over: a cell (blue) receives transcripts from its neighbours in the pruned Delaunay graph, in proportion to the kernel weights β_{ij} (arrow widths); the niche descriptor $\mathbf{c}_i$ reads the neighbours’ types and the surrounding tissue image, with the cell itself masked out. **b**, The generative model (Equations (1) to (6)): the intrinsic state $\mathbf{z}_i$ and the response $\mathbf{w}_i$, whose prior is set by the niche and the type, decode to the cell’s own composition $\boldsymbol{\rho}_i$, which is mixed with the neighbours’ compositions at the fixed fraction κ . An adversary keeps the niche out of $\mathbf{z}_i$ given the type. **c**, Why κ is swept (Proposition 1): on the simplex of three genes, each assumed κ implies a clean composition on a line running from the observed composition $\mathbf{p}_i$ away from the neighbours’ influx $\bar{\boldsymbol{\rho}}_i$; every κ fits until a gene’s share would turn negative at $\bar{\kappa}_i$, so the data bound κ from above only. Schematic. **d**, The breakdown point (Definition 1): a finding is refitted at every κ of the grid, and κ^* is the smallest κ at which its interval reaches its null value. Schematic.

Spill-over. The influx $\bar{\boldsymbol{\rho}}_i$ is the neighbours’ clean compositions averaged with a fixed kernel, $\beta_{ij} \propto f_{ij} \exp(-d_{ij}/\tau)$, normalised over $\mathcal{N}_i$, where d_{ij} is the distance between centroids, f_{ij} the length of the shared Voronoi face and $\tau = 20 \mu\text{m}$. The kernel is computed once and not learned, unlike the per-cell neighbour weights of resolVI (Ergen and Yosef, 2025). The mixture of Equation (5) is exact if transcript sources are independent Poisson variables and the spilled-over rate scales with the receiving cell’s own rate; the multinomial then follows from conditioning on the observed total. The spill-over it represents has a specific form: one hop, one fraction for the whole section, and no gene dependence. An isolated cell receives none (Supplement S1.1).

2.2 Inference and objective

Both latents have amortised diagonal-Gaussian posteriors. The encoder of $q(\mathbf{z}_i \mid \mathbf{x}_i, t_i)$ reads the cell’s library-normalised log counts, its log total and its type, and never the niche. The posterior $q(\mathbf{w}_i \mid \mathbf{c}_i, t_i, \mathbf{z}_i, \mathbf{x}_i)$ reads the niche and also the cell, since $\mathbf{z}_i$ and $\mathbf{x}_i$ together carry evidence about $\mathbf{w}_i$ that the niche alone does not. Writing $r_i(\mathbf{w}) = \ell_i^{-1} \sum_g x_{ig} \log p_{ig}(\mathbf{z}_i, \mathbf{w})$ for the reconstruction per count, the training objective, maximised, is

$$J = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \left[r_i(\mathbf{w}_i) + \omega r_i(\tilde{\mathbf{w}}_i) - (1 + \omega) \alpha_z \text{KL}(q(\mathbf{z}_i) \parallel p(\mathbf{z}_i \mid \mathbf{x}_i, t_i)) - \alpha_w \text{KL}(q(\mathbf{w}_i) \parallel p(\mathbf{w}_i \mid \mathbf{c}_i, t_i)) \right] - \alpha_a \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \text{Adv}_i \quad (8)$$

over a batch $\mathcal{S}$ of cells. With $\omega = 0$ and $\alpha_a = 0$ this is an evidence lower bound, scaled by depth so that the weights transfer between sections of different depth. The second term is the *intrinsic path*: the same decoder evaluated with $\tilde{\mathbf{w}}_i$ drawn from the prior instead of the posterior, a second bound on the same evidence (Slavutsky et al., 2026) that asks $\mathbf{z}_i$ to explain the cell given only the response *expected*

in its niche. It keeps the type out of $\mathbf{w}$: refitted with $\omega = 0$, the response’s agreement with type rises on all four sections (Supplement S8.2). The last term is the adversary of Section 2.3. The neighbours’ compositions $\boldsymbol{\rho}_j$ in $\bar{\boldsymbol{\rho}}_i$ enter under a stop-gradient, as in resolVI (Ergen and Yosef, 2025), so that a neighbour’s profile is not trained to absorb the cell’s residual; batches are contiguous spatial tiles with a two-hop halo (Supplement S3.3). We use $d_z = 20$, $d_w = 6$, $\omega = 1$, $\alpha_z = \frac{1}{2}/\bar{\ell}$ with $\bar{\ell}$ the section’s median total, and $\alpha_w = 0.1$; the remaining settings are in Supplement S3.1.

2.3 Conditional invariance

The architecture keeps the niche out of the encoder of $\mathbf{z}$, but not out of the counts it reads. The target is that, within a type, the intrinsic state says nothing about the niche:

$$\mathbb{I}(\mathbf{z}_i; [\mathbf{y}_i, \Phi_i] \mid t_i) = 0. \quad (9)$$

It is conditional on the type because cell types cluster in space: “I am a T cell” predicts “my neighbours are T cells”, and an intrinsic state blind to the niche unconditionally would have to discard its type (?). It covers the image as well as the composition, so that a niche defined by morphology alone (a vessel, fibrosis) cannot hide in $\mathbf{z}$. Two heads with parameters $\mu_{z,i}$ and $\bar{\mu}_{z,i}$ predict, from the posterior mean $\mu_{z,i}$ and the type, the neighbour-type composition $\mathbf{y}_i$ and a soft membership e_i^Φ of Φ_i in K image niches; the encoder is penalised by their skill beyond knowing the type alone,

$$\text{Adv}_i = \lambda_y [\text{CE}(\mathbf{y}_i, \bar{\mathbf{y}}(t_i)) - \text{CE}(\mathbf{y}_i, \hat{\mathbf{y}}_\xi(\mu_{z,i}, t_i))] + \text{CE}(e_i^\Phi, \bar{\Phi}(t_i)) - \text{CE}(e_i^\Phi, \hat{e}_\xi^\Phi(\mu_{z,i}, t_i)), \quad (10)$$

where $\bar{\mathbf{y}}(t)$ and $\bar{\Phi}(t)$ are the mean targets of type t and $\lambda_y = 3$ weights the composition, the channel of the spill-over confound. The heads maximise Adv_i

with their own optimiser, the encoder minimises it, and it is zero at the optimum. Type-conditional adversarial invariance of an intrinsic latent against neighbour-type composition was introduced by Mint-Flow’s per-type critics (Akbarnejad et al., 2025); what we add is the image target and an independent test of the result.

The held-out probe. A training adversary that is defeated proves nothing (?), so invariance is graded by a *fresh* predictor of each target block b (the composition, and the leading principal components of Φ) from $(\mu_{z,i}, t_i)$, trained on spatially separate training tiles and scored on held-out tiles. Its held-out gain over the type mean $\bar{v}_k(t)$, per component k of the block,

$$G_b = \frac{1}{|b|} \sum_{k \in b} \frac{1}{2} \log \frac{\mathbb{E}(v_k - \bar{v}_k(t))^2}{\mathbb{E}(v_k - \hat{v}_k(\mu_z, t))^2}, \quad (11)$$

estimates the predictive information from $\mathbf{z}$ to the block given the type (?). Its value with μ_z permuted among cells of the same type is the chance floor. The **residual niche signal** is the excess of G_b over that floor, reported as $1 - e^{-2 \text{excess}}$: the share of the block’s within-type variance that the probe explains beyond chance. We fit a linear (ridge) and a nonlinear (small perceptron) probe, each against its own floor, and compare with a fit without the adversary ($\alpha_a = 0$). A probe at its floor bounds the dependence it can detect; it does not certify independence. Checkpoints are selected on held-out reconstruction subject to the agreement of $\mathbf{z}$ with type (normalised mutual information of a k -means clustering of μ_z with t) staying within 90% of its running maximum, so that reconstruction is never bought by emptying $\mathbf{z}$ (Supplement S3.1).

2.4 What is identified

Spill-over and response both make a cell resemble its neighbours, and a flexible model can fit either. The counts constrain the spill-over fraction in one direction only. Removing more and more of the neighbours’ profile from a cell eventually drives some gene’s share below zero, which no clean profile can have: this bounds κ from above. Every smaller κ yields a valid clean profile that reproduces the data exactly, so there is no lower bound (Figure 1c).

Proposition 1. *Hold the neighbours’ clean compositions, and hence $\bar{\rho}_i$, fixed, and let $\mathbf{p}_i = (1 - \kappa)\rho_i + \kappa\bar{\rho}_i$ with ρ_i in the simplex. A fraction $\kappa' \in [0, 1)$ reproduces $\mathbf{p}_i$ with a clean composition in the simplex if*

and only if $\kappa' \leq \bar{\kappa}_i = \min_{g: \bar{\rho}_{ig} > 0} p_{ig}/\bar{\rho}_{ig}$. The true κ satisfies $\kappa \leq \bar{\kappa}_i$, with equality if and only if $\rho_{ig} = 0$ for some gene g with $\bar{\rho}_{ig} > 0$.

The proof is in Supplement S2.6. Even with the neighbours’ clean profiles known, then, the counts bound κ only from above, and pin it only through a gene that the cell cannot express itself. With finite counts the bound is soft, and we do not compute it on our sections. In simulation, recovery of planted states and programmes barely changes with the assumed κ (Section 3.2), which is the same non-identifiability seen in practice. Given κ , the response is identified at most up to a rotation of its programmes and a per-type offset, so we report only rotation-invariant summaries and within-type contrasts (Supplement S1.5).

The sweep. We therefore do not estimate κ . We fit the model independently at each $\kappa \in \{0, 0.05, 0.1, 0.2, 0.3, 0.4\}$, with three seeds per value and every other weight held fixed, and check the held-out probe at each value, so that a change in a readout is not a change in the invariance. This is a sensitivity analysis with κ as the sensitivity parameter, in the tradition of tipping-point and E-value analyses of unmeasured confounding (Rosenbaum, 2002; ?): the fractions compatible with the data form an interval from zero (Manski, 2003), and the question asked of each finding is how far into it the finding survives.

Definition 1 (Breakdown point). For a signed contrast r , let $I_\kappa(r)$ be the interval of its mean over seeds at spill-over fraction κ . The contrast holds at κ if $I_\kappa(r)$ excludes zero and every seed keeps the sign the contrast has at $\kappa = 0$. Its *breakdown point* $\kappa^*(r)$ is the smallest grid value at which it does not hold. If it does not hold at $\kappa = 0$, r is no finding; if it holds at every grid value, $\kappa^*(r)$ is reported as exceeding the top of the grid.

The interval comes from a spatial block bootstrap over tiles, Bonferroni-adjusted over the contrasts claimed on a section. A finding with $\kappa^* > 0.4$ cannot be explained by spill-over of the modelled form up to four times the typical misassignment of about a tenth of a cell’s transcripts (Yang et al., 2025); spill-over of another form (longer-ranged, gene-specific, depth-dependent) is not ruled out. Readouts shrink as κ grows by construction, since more of the neighbours’ resemblance is assigned to the mixture, so only the sign and the interval are read, not the size. A contrast without a sign at $\kappa = 0$, because it is zero there by construction or measures decoding rather than spill-over removal, is shown as a trajectory instead (Supplement S8.1). Nothing in the protocol is specific

to DISCELL: any model with a spill-over term at a fixed fraction can attach a breakdown point to any spatial readout.

3 Experiments

3.1 Setup

[TODO: setup]

3.2 Simulation

[TODO: simulation]

3.3 What the Intrinsic State and the Response Carry

[TODO: intrinsic state and response]

3.4 What the Response Encodes

[TODO: programme atlas]

3.5 Comparison with Other Methods

[TODO: comparison]

3.6 Relocation

[TODO: relocation]

3.7 Robustness to Spill-over

[TODO: robustness]

4 Discussion

[TODO: discussion]

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